Application:
What Does a Year of College Cost?

What you give up	How it adds up	One year
Tuition and fees	UC tuition plus campus fees	$17,000
Books and supplies	roughly, over three quarters	$2,000
Forgone earnings	a year of full-time pay you did not earn	$48,000
Total, one year	**out of pocket + opportunity cost**	**$67,000**

Not counted: housing and food, about $21,000. You have to live and eat either way — so only the difference between the dorm and the alternative is a cost of going.

The bill says $19,000. The cost is $67,000 — because the real price of a degree is the years, not the tuition.

And What Does It Return?

Median full-time pay: about $1,500 a week with a degree, about $900 a week without.

$$\$1{,}500 - \$900 = \$600 \text{ a week}$$

$$\$600 \times 52 \text{ weeks} \approx \$30{,}000 \text{ a year}$$

$$\$67{,}000 \times 4 \text{ years} = \$268{,}000$$

$$\$268{,}000 \div \$30{,}000 \text{ a year} \approx 9 \text{ years}$$

Nine years of work after graduating, and the degree has paid for itself.

Chapter 3

Optimization: Trying to Do the Best You Can

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 1 gave you three principles. Chapter 3 takes the first one apart.

TRADE-OFF

Choosing one thing means giving up another.

There is no option that costs nothing.

BUDGET CONSTRAINT

The set of things you can actually choose.

Money, hours, attention — all of them bind.

OPPORTUNITY COST

The value of the best alternative you gave up.

This is the one we will put a dollar sign on today.

Chapter 1 said people try to pick the best feasible option. Chapter 3 shows you how to actually find it.

Today's Roadmap

01

What Optimization Is

The best feasible option — and why “feasible” is the load-bearing word

02

Optimization Using Total Value

Convert to common units, add it all up, pick the best number

03

Optimization Using Marginal Analysis

Compare the change between options, not the level of each one

04

Evidence, and Four Problems

Does distance really price housing? Then practice, with a warning attached

PART 1

Choosing the Best Feasible Option

Optimization, Defined

Optimization is choosing the best feasible option, given the information available.

THE WORDS TO KEEP

Economists call the best feasible option the optimal choice, or simply the optimum.

Best out of the menu you face — not the menu you wish you faced.

WHAT THE CLAIM IS NOT

Not that people always succeed.

The claim is that people try, and that trying is systematic enough to model — and to predict.

Economists use optimization to predict most of the choices that people, households, firms and governments make.

“Feasible”

Three constraints, all of them from Chapter 1, all of them still binding.

MONEY

$1,200 a month for rent.

The $2,000 loft is not a worse option. It is not an option at all.

TIME

Twenty-four hours a day.

Two things you would enjoy at 8 p.m. — you get one of them, and only one.

AVAILABILITY

The 9 a.m. section is full.

Your optimum is defined over the seats that are left, not the seats you wanted.

Change the constraint and the optimum changes — without anybody changing what they want.

Do People Always Optimize?

No. Optimization is a model of behaviour, not a claim that everybody gets it right.

LIMITED INFORMATION

Information is costly to get, so you decide on less than all of it.

Nobody reads every review



TOO MUCH INFORMATION

Sometimes there is so much of it that you cannot tell where to start sorting.



INEXPERIENCE

The first time you face a situation you cannot use the information you do have.

Rookie mistakes are real.



Behavioral economics is the branch that studies exactly where this breaks down — procrastination, self-control, addiction.

When the Model Earns Its Keep

WHERE OPTIMIZATION PREDICTS WELL

Choices people make repeatedly.

- Weekly grocery shopping
- The route to work
- A firm pricing the same product for the tenth year running

Practice is the mechanism: mistakes get corrected because the situation comes round again.

WHERE IT PREDICTS BADLY

Choices people make once.

- Your first poker hand
- Choosing a mortgage
- Picking a major at eighteen

No feedback loop, no correction — and often the stakes are highest here.

The useful question is never “are people rational?” — it is “is this a situation where the model earns its keep?”

Two Techniques

net benefit = total benefits − total costs

marginal net benefit = change in benefits − change in costs

OPTIMIZATION USING TOTAL VALUE

Compute the net benefit of every option. Pick the biggest number.

Works on levels: what is this option worth, all in?

OPTIMIZATION USING MARGINAL ANALYSIS

Compute the change in net benefit when you move from one option to the next. Keep moving while it helps.

Works on differences.

They always give the same answer. Use whichever is easier for the problem in front of you.

What “Marginal” Means

Marginal means the change — what happens when you move from one alternative to the next.

LEVELS

Total cost of Close: $1,190
Total cost of Far: $1,180

.

DIFFERENCES

Marginal total cost of moving
from Close to Far: $1180-$1190=−$10

Marginal analysis is usually faster, because everything the two options have in common cancels out and never gets written down.

“Need More Snooze”



Photo: BitsAndSplits / Getty Images

EVERY SNOOZE, FROM THE FIRST TO THE TENTH

marginal cost	**5 minutes**
marginal benefit	**5 minutes of sleep**

Ten snoozes later, the next one still costs five minutes. The comparison is identical every time — so nothing inside it ever tells him to stop.

The Second Chip and the Second Slice



CHIPS

Every chip is about as good as the last, so marginal utility barely falls. Nothing says stop until you notice the bag is empty.

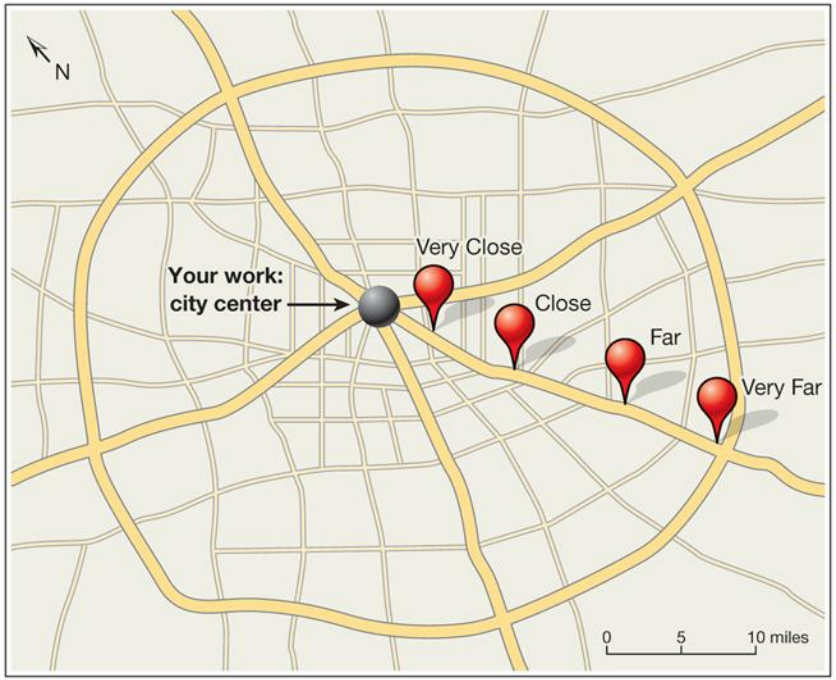
PIZZA

The second slice is worth much less than the first, and by the third you are full. Marginal utility falls fast — and it tells you when to stop.

PART 2

Optimization Using Total Value

The Problem: Where Should You Live?



Acemoglu, Laibson & List, Microeconomics 3e

WHERE SHOULD YOU LIVE?

Your job is in the city centre. Four apartments, identical except for two things: the commute and the rent.

Rent falls as you move out. Commuting time rises.

Exhibit 3.1 — Your Short List

Four apartments, differing only in commuting time and rent.

Apartment	Commuting time (hours / month)	Rent ($ / month)
Very Close	5 hours	$1,180
Close	10 hours	$1,090
Far	15 hours	$1,030
Very Far	20 hours	$1,000



What Does It Cost to Commute?

OUT OF POCKET

Fuel. Transit fares. Parking. Tolls.

Real money, easy to measure — and we are going to ignore it.

WEAR AND TEAR

The car depreciates faster the more you drive it.

Real, and almost nobody counts it.

YOUR TIME

What you would have to be paid to sit in traffic instead of doing something else.

This is the one we price.

Assume an opportunity cost of time of $10 per hour

Optimization Using Total Value: Three Steps

01

Translate everything into common units

Here, dollars per month.

02

Compute the total net benefit of each alternative

Total benefits minus total costs, for every option on the list.

03

Pick the alternative with the highest net benefit

Exhibit 3.2 — Total Cost at $10 an Hour

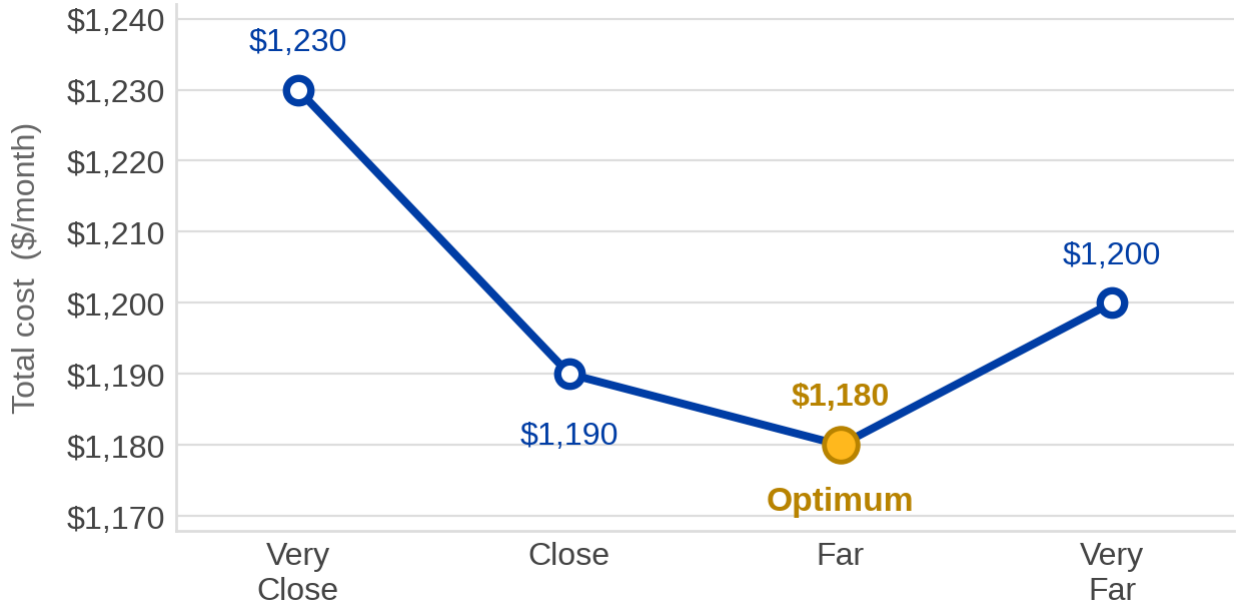
Apartment	Commuting time (hours / month)	Commuting cost ($ / month)	Rent ($ / month)	Total cost ($ / month)
Very Close	5	$50	$1,180	$50+$1,180=$1,230
Close	10	$100	$1,090	$1,190
Far	15	$150	$1,030	$1,180
Very Far	20	$200	$1,000	$1,200

Very Far: 20 hrs × $10 = $200, + $1,000 = $1,200

Far: 15 hrs × $10 = $150, + $1,030 = $1,180

Rent alone says Very Far. Total cost says Far. The cheapest rent is not the cheapest apartment.

Exhibit 3.3



The curve is U-shaped because two costs move in opposite directions: rent falls as you move out, commuting cost rises.

Exhibit 3.4 — Total Cost at $15 an Hour

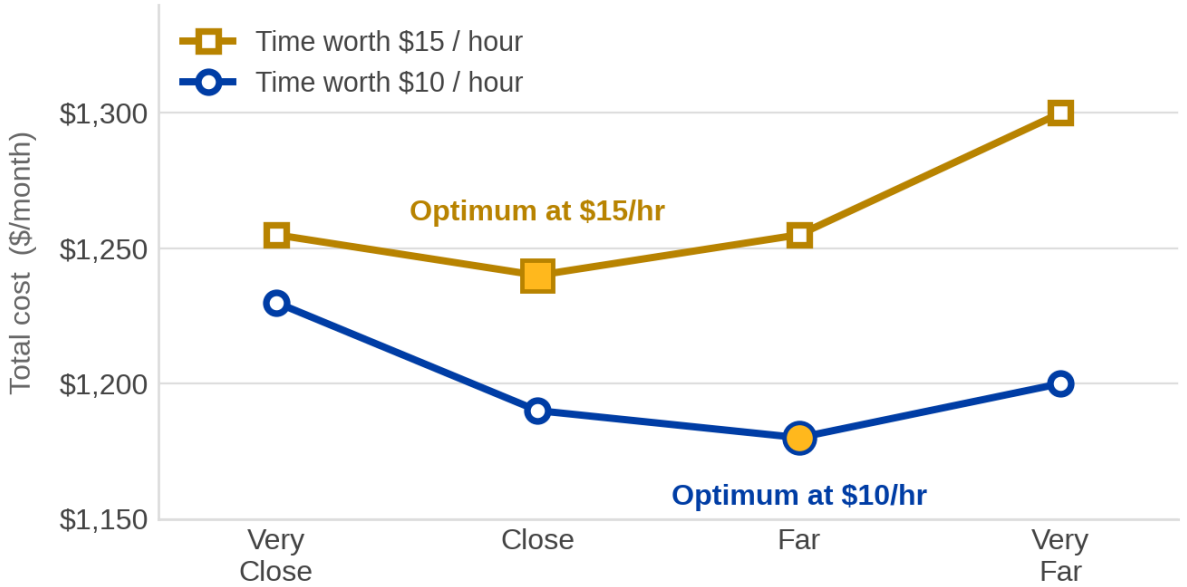
Apartment	Commuting time (hours / month)	Commuting cost ($ / month)	Rent ($ / month)	Total cost ($ / month)
Very Close	5	$75	$1,180	$75+$1,180=$1,255
Close	10	$150	$1,090	$1,240
Far	15	$225	$1,030	$1,255
Very Far	20	$300	$1,000	$1,300

Far was $1,180 at $10/hr. Now $1,255.

Close was $1,190. Now $1,240 — and it wins.

The optimum moved one step closer to the centre. Nobody's preferences changed — one price did.

Exhibit 3.6 — Both Curves Together



People whose time is worth more should live closer in. That is a prediction about the world — and Part 4 checks it against data.

PART 3

Optimization Using Marginal Analysis

Marginal Analysis in Three Steps

01

Translate everything into common units

Dollars per month.

02

Compute the marginal consequences of moving between alternatives

For each step from one option to the next: what changes, and by how much?

03

Choose the alternative you would not leave in either direction

Moving to it makes you better off; moving away from it makes you worse off.

Step 3 is the whole idea. An optimum is a place you would not want to walk away from — forwards or backwards.

Exhibit 3.7 — Cost and Marginal Cost

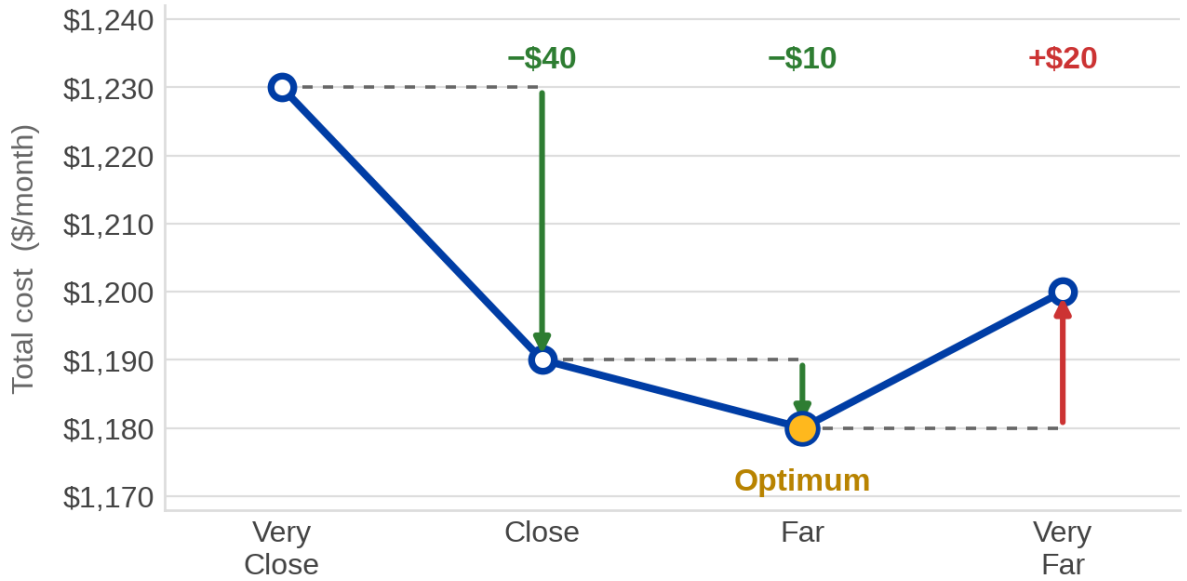
Still at an opportunity cost of time of $10 an hour. Marginal columns read “change from the row above.”

Apartment	Commuting cost	Marginal commuting cost	Rent	Marginal rent	Total cost	Marginal total cost
Very Close	$50	—	$1,180	—	$1,230	—
Close	$100	+$50	$1,090	−$90	$1,190	−$40
Far	**$150**	**+$50**	**$1,030**	**−$60**	**$1,180**	**−$10**
Very Far	$200	+$50	$1,000	−$30	$1,200	+$20

Read the last column: −$40, then −$10, then +$20.

Keep moving while the number is negative. Stop the step before it turns positive.

Exhibit 3.8 — Marginal Cost as a Picture



Every arrow is the marginal total cost of one move. The optimum is the last point you reach before an arrow starts pointing up.

The Principle of Optimization at the Margin

An optimal feasible alternative has the property that moving to it makes you better off, and moving away from it makes you worse off.

MOVING TO FAR

From Close:

pay $50 more commuting,
save $60 on rent —
so you are $10 better off. Go.

MOVING AWAY FROM FAR

On to Very Far:

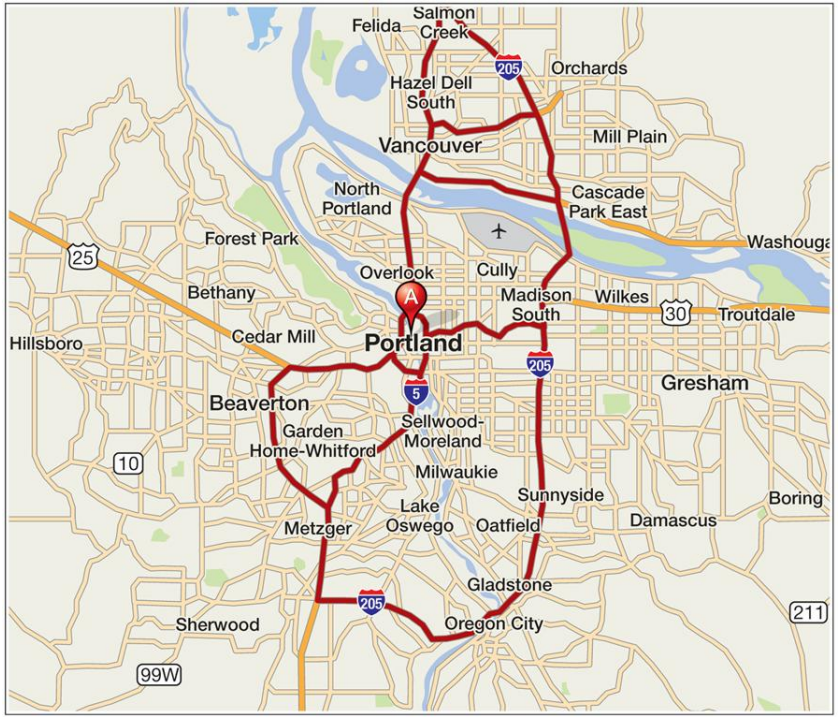
pay $50 more commuting,
save only $30 on rent —
so you are $20 worse off. Stay.

Total value and marginal analysis picked the same apartment. They always will.

PART 4

Evidence: Does Location Really Price Housing?

Why Portland's Curve Has a Shelf



Acemoglu, Laibson & List, Microeconomics 3e

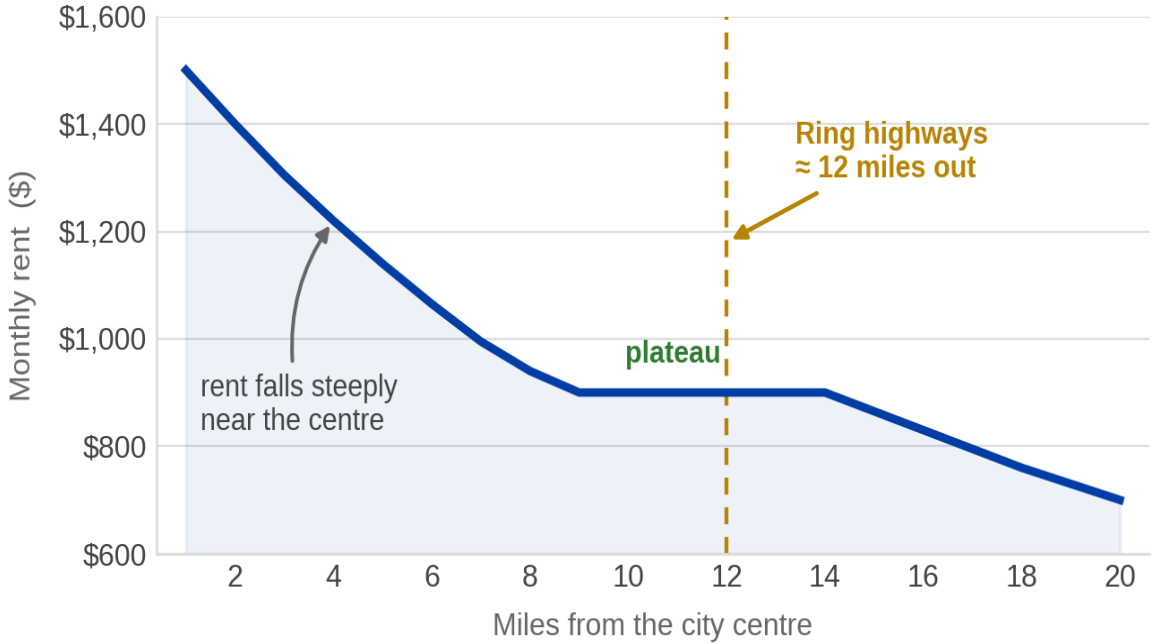
THE RING ROADS

Portland is circled by highways about twelve miles out from the center.

They cut commuting time for the areas they serve — so those apartments are effectively closer to downtown than their mileage suggests.

The model was written in commuting hours, not miles. Watch what that does to the rent curve on the next slide.

Exhibit 3.9 — Rent and Distance in Portland

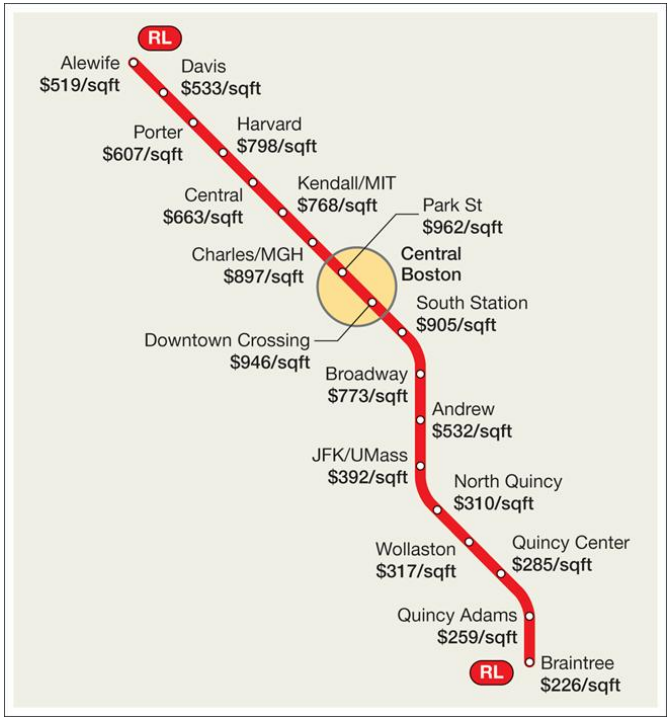


Rents fall about 33% between the city center and identical apartments six miles out.

WHY THE PLATEAU?

The ring highways cut commuting time for the areas they serve — so those flats are closer than the mileage.

The Same Thing, One Subway Line



PARK STREET IS THE CENTRE

Price per square foot peaks at Park Street, in the heart of downtown Boston, and falls in both directions along the line.

Alewife, north end: $519
Park Street: $962
Braintree, south end: $226

Distance from the centre is counted in stops, not miles. Harvard at $798 is the exception — a second centre in its own right.

Practice: Two Apartments in Portland

You want to pay about $950 a month near the ring road. You have found two apartments, both at $950.

APARTMENT EAST

$950 per month.

A great view of Mt. Hood.

Further from the airport.

APARTMENT WEST

$950 per month.

No view.

Thirty minutes nearer the airport.

Four questions. Each one changes exactly one number — watch how little it takes to flip the answer.

Question 1 — The View

East has a view of Mt. Hood and West does not. You value that view at $25 a month. Which do you rent?

TOTAL VALUE

Call the housing itself worth V to you — the apartments are identical, so V is the same for both.

East: V + $25 − $950
West: V + $0 − $950
V cancels. East is ahead by $25.

MARGINAL ANALYSIS

West → East: +$25

East → West: −$25

Moving to East helps; leaving hurts.

Rent Apartment East.

Question 2 — The Airport, Twice a Month

West is 30 minutes nearer the airport. You fly twice a month — four drives. Your time is worth $20 an hour. Which do you rent?

TOTAL VALUE

The extra driving: 4 × 30 min = 2 hours,
at $20 an hour = $40 a month.

East: V + $25 − $950 − $40
West: V + $0 − $950
West is ahead by $15.

MARGINAL ANALYSIS

West → East:
+$25 view − $40 driving = −$15

East → West: +$15

Moving to East hurts. Stay.

Rent Apartment West.

Questions 3 and 4 — Less Travel, Then a Raise

Q3 YOU NOW FLY ONCE A MONTH

Two drives, not four — one extra hour.
1 hour × $20 = $20

TOTAL VALUE
East: V + $25 − $950 − $20
West: V + $0 − $950

MARGINAL
+$25 view − $20 driving = +$5

→ **East, ahead by $5.**

Q4 AND THEN YOU GET A RAISE

Still one extra hour, but your time
is now worth $23. 1 × $23 = $23

TOTAL VALUE
East: V + $25 − $950 − $23
West: V + $0 − $950

MARGINAL
+$25 view − $23 driving = +$2

→ **Still East — by $2.**

Chapter 3: The Full Picture

	What it says	Key terms	Where you saw it
Optimization	People choose the best feasible option, given the information they have	optimal choice, optimum, feasible	Four apartments, one job downtown
Total value	Convert to common units, compute net benefit for every option, take the highest	net benefit, common units	Far wins at $1,180 a month
Marginal analysis	Compare the change between options, not the level of each; move while moving helps	marginal cost, marginal benefit	−$40, then −$10, then +$20
At the margin	An optimum is a point you would not leave in either direction	optimization at the margin	Far, checked both ways

Four rows. Both methods, on every problem, always agree.

KEY TAKEAWAYS

Chapter 3: Optimization

- Optimization means choosing the best feasible option with the information you have — not always getting it right
- Convert everything to common units first. You cannot compare hours with rent until the hours have a price
- Total value: compute the net benefit of every option, and take the highest
- Marginal analysis: compute the change between options, and keep moving while the change helps
- Both always agree — and an optimum is a point you would not leave in either direction